# Mengmeng Wang

*Data Scientist, Graduate Researcher*

+61-466-989-626
mengmengw@student.unimelb.edu.au
/mengmwang
/mengmwang
/mengmwang.github.io

## Profile

I am a dedicated data scientist & statistical signal processing researcher with a background in biomedical engineering and computer science. I enjoy solving challenging problems in data simulation, statistical data modelling and data analytics. I have developed time-series analysis, predictive modelling, data visualisation & analysis, machine learning and image processing algorithms. I am experienced in Python, R, MATLAB, Excel, Power BI, Tableau, and Pandas, NumPy, Scikit-Learn, NLTK, SciPy, Matplotlib, JSON, RegEx libraries.

## Skills

**Programming Languages** Python, Jupyter Notebook, R, SQL, MATLAB

**Technical Tools** Tableau, Power BI, GitHub, Excel

**Python Libraries** Pandas, NumPy, Scikit-Learn, NLTK, SciPy, Matplotlib, JSON, RegEx

**Professional** Data Analysis, Technical Writing, Project Management, Teamwork, Communications

## Education

**Doctor of Philosophy – Statistical Signal Processing** 2017 – 2022 (Expected)
*The University of Melbourne*

**Master of Science – Image Processing** 2013 – 2014
*University of Bristol*

**Bachelor of Engineering – Telecommunications Engineering** 2009 – 2013
*Beijing University of Posts and Telecommunications*

## Experience

**Graduate Researcher** April 2017 – Present
*The University of Melbourne*

- PhD Project: signal analysis techniques in resting state functional near-infrared spectroscopy.
- Applied quantitative mathematical modellings to model functional brain activities.
- Investigated the impact of noise on brain signals and analysed brain functional connectivity using signal processing and statistical tools.
- Developed 3+ statistical algorithms from pure mathematical derivations to MATLAB implementations.

**Data Processing & Machine Learning Tutor (Casual)** March 2019 – Present
*The University of Melbourne*

- Tutored 3 university-level subjects: elements of data processing, machine learning, signals and systems.
- Key topics: data wrangling, format, visualisation, natural language processing, supervised and unsupervised learning, neural network.
- Used Python and Pandas, NumPy, Scikit-Learn libraries to teach modern data science and advanced machine learning concepts.

**Consulting Engineer** May 2015 – April 2017
*China Academy of Information and Communications Technology*

- Analysed domestic and international technologies in telecommunications and information science.
- Provided strategic advice on markets and created technical models and evaluation tools.
- Produced over 20 consulting reports and data analysis deliverables to governmental and industrial clients.

## Selected Projects (Available on GitHub)

### EEG Data Analysis in Music Therapy

- Performed EEG Data Analysis and visualisations in music therapy research.
- EEG Data importing, cleaning, preprocessing, feature engineering, visualisations and statistical analysis.
- Collaborated with health professionals, doctors and music therapists, to investigate the impacts of music therapy on brain state and functional brain connectivity.
- Tools used: MATLAB and EEGLAB Toolbox.

### Financial Timeseries Processing and Forecast

- Performed financial data analysis in Level 1 Limit Order Book data, across data processing pipeline.
- Data cleaning and preparation: outlier detection, data visualisation and feature engineering.
- Data analysis: correlation, moving-average, auto-regression analysis.
- Timeseries forecasting: auto-regression model and machine learning models (decision tree, logistic regression, neural networks).
- Tools used: Python, Jupyter Notebook, Pandas, NumPy, matplotlib, seaborn, statsmodels, sklearn, keras.

### Customer Purchasing Behaviours Analysis

- Analysed customer transaction and purchase behaviour data to find patterns in customer behaviours.
- Perform data cleaning, preparation and visualisation to facilitate analysis.
- Draw insights on change in store layout in customer behaviours.
- Tools used: Python, Jupyter Notebook, Pandas, NumPy.

### Reinforcement Learning and Multi-Armed Bandits (MABs)

- Implementation and evaluation of state-of-art reinforcement learning algorithms: $\epsilon$-Greedy, Upper Confidence Bound (UCB), LinUCB and KernalUCB.
- Tools used: Python, Jupyter Notebook, NumPy.

## Publications

- **M. Wang**, C. Davey and L. Johnston, "Correction of induced functional connectivity in filtered resting state fNIRS data," *The 27th Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, 2021.
- **M. Wang**, L. Johnston and C. Davey, "Correction for time-varying signal power in fNIRS connectivity analyses," *Society of fNIRS Virtual Conference*, 2021.
- **M. Wang** and A. Seghouane, "Motion Artefact Removal in Functional Near-infrared Spectroscopy Signals Based on Robust Estimation," *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2019.
- **M. Wang**, F. Zhang and D. Agrafiotis, "A very low complexity reduced reference video quality metric based on spatio-temporal information selection," *IEEE International Conference on Image Processing*, 2015.

## Certificates and Awards

- Online Course Certificates in Data Science 101, Data Science Tools, Big Data 101, Introduction to Cloud, Blockchain Essentials
- HackerRank Programming Certificates for Python and SQL
- Virtual Internship Certificates from ANZ Australia, Goldman Sachs, JPMorgan Chase & Co.
- Virtual Internship Certificates from KPMG Australia, Deloitte Australia, Quantium, General Electric
- Conference Travel Scholarships, The University of Melbourne
- Melbourne Research Scholarships, The University of Melbourne
- Excellent Examination Performance Prize, University of Bristol

## Volunteer Experience

**Girl Power Mentor** March 2021 – Present

*The University of Melbourne*

- Mentored Year 11/12 female students with interests in science and engineering.
- Coached girls to develop strength and confidence to pursue tertiary education in science and engineering.

## References

Available Upon Request